

# India Research Corps



## Thinking like a Researcher

*Induction Modules for IRC Scholars*

## MODULE ONE: THE SCIENTIFIC METHOD

*How Should a Researcher Think?*

What makes scientific enquiry *scientific*?

## Can anecdotal observation be scientific research?

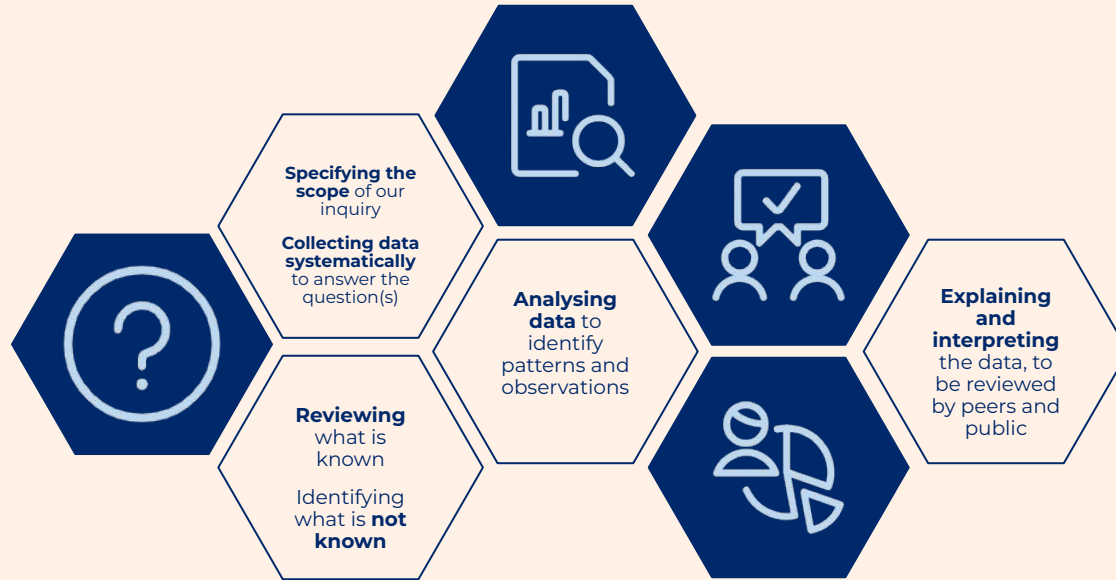
### Anecdotal Observation

- Often subjective, prone to observer bias
- Measurement differs between observers & situations
- Observations may be different in different instances
- No consensus on logic and rules of observation and analysis
- Does not need to be extrapolated beyond its immediate context

### Scientific Research

- **Objectivity** is extremely important
- **Validity** (accurate measurement of the true properties of the real world) is important
- **Reliability** (consistency of results under the same conditions) is a central tenet
- **Transparency** of methods, analysis, and interpretation is expected
- **Replicability** is built into the research

The scientific process is rooted in these central tenets.



**Objectivity | Validity | Reliability | Transparency | Replicability**

# Exercise

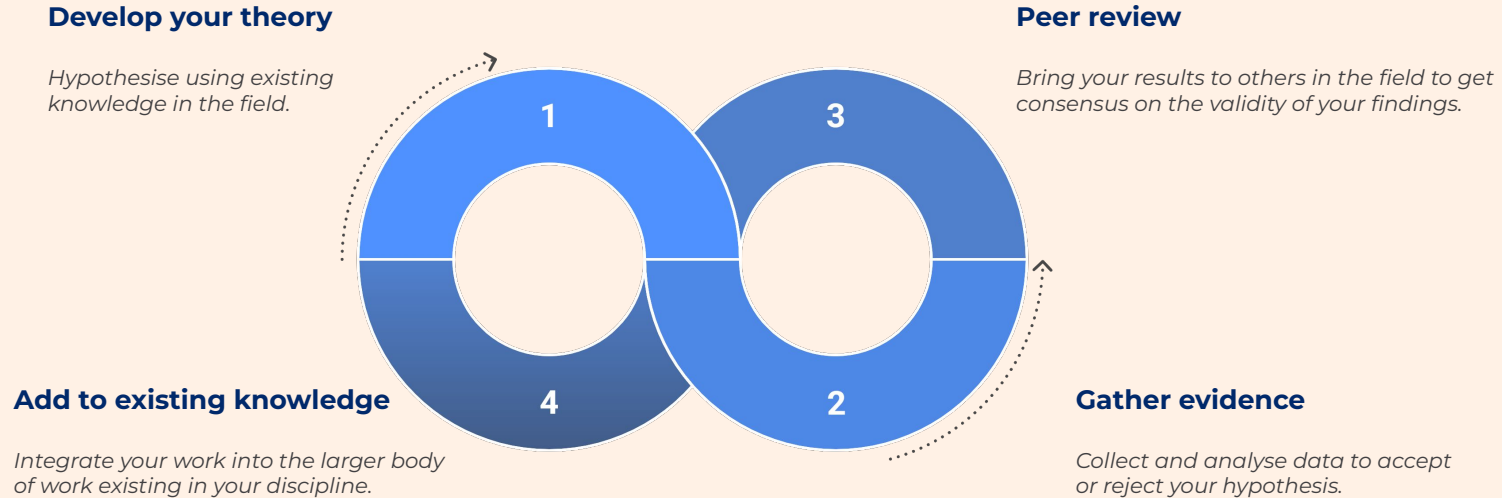
*Think of the way you have worded your research question, the methodology you have selected, and your hypotheses.*

*Score yourself out of 5 on the following criteria:*

1. **Objectivity:** Is your work unbiased, evidence-based, and free from personal influence?
2. **Validity:** Are your concepts, methods, and tools actually measuring what they intend to measure?
3. **Reliability:** If repeated under the same conditions, would your study produce similar results?
4. **Transparency:** Are you openly sharing your methodological choices, data, and analysis process so that your peers can evaluate, replicate, and build upon your work?
5. **Replicability:** If your peers tried to answer the same question—either using your data & methods, or with their own—would they obtain similar results?

*Where are you scoring the lowest? What are some ways you can improve your score on that criterion?*

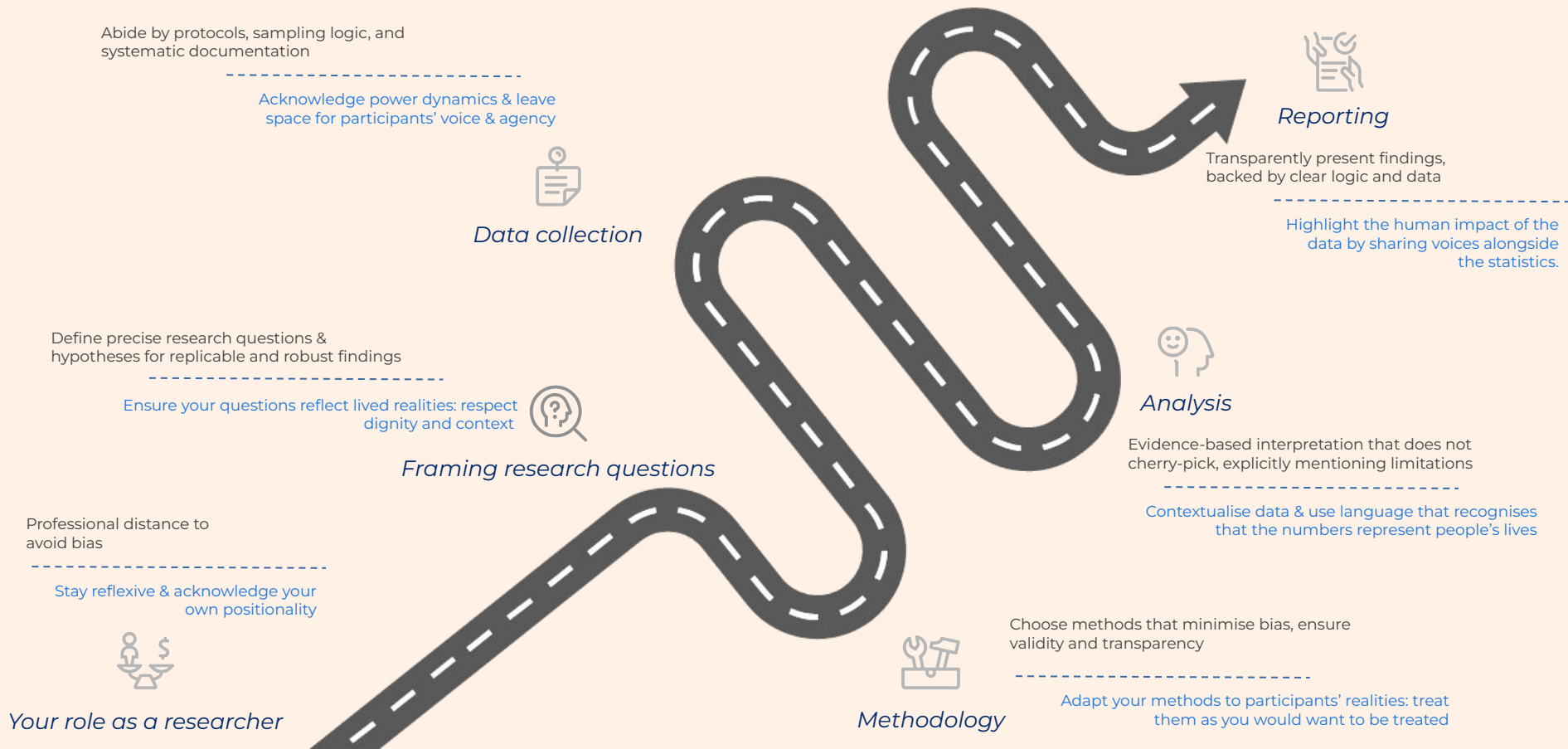
Science requires you to converse and engage with others.



**Scientific consensus** is vital for establishing theory; **falsifiability** of established theories is equally important.

**Theory, evidence, and peer review** are what tie your work into the larger body of knowledge draw from and ultimately hope to contribute to.

# How does one balance scientific rigour & deep empathy as a researcher?





# Thinking exercise

*Critically examine your own work on your research question until now. As a researcher, what do you currently favour more: **scientific rigour** or **deep empathy**?*

*List three changes you think you need to make to your approach from here on out to address this imbalance.*

# Recommended readings

1. Bhattacharjee, Anol. *Social Science Research: Principles, Methods, and Practices* (2012). Textbooks Collection. 3. [URL](#).
2. “Guiding Principles for Scientific Inquiry”, *Scientific Research in Education*, 2002, The National Academies Press, pp 50-79. [DOI](#).
3. *How do we know what’s true?*, Humanists UK, YouTube, [URL](#).
4. “The principal elements of the nature of science: dispelling the myths” Adapted from the chapter in W. F. McComas (ed.) *The Nature of Science in Science Education*, 1998, Kluwer Academic Publishers, pp. 53-70. [URL](#).

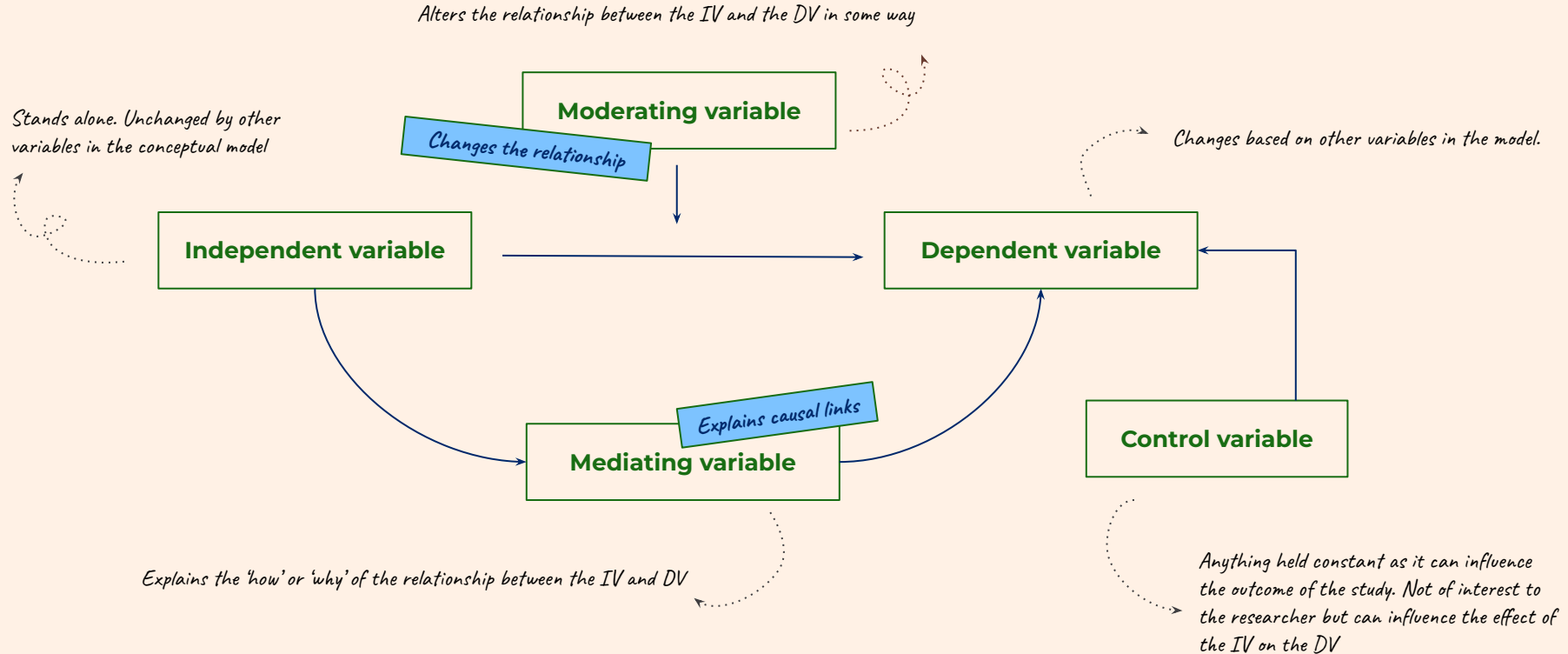
## **MODULE TWO: BUILDING A CONCEPTUAL MODEL**

*What Do You Think is Going On?*

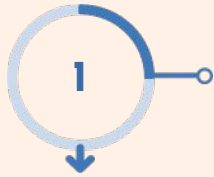
A conceptual model or framework can guide research by providing a visual representation of theoretical constructs and variables of interest: what do you think is going on?

*Mind maps, diagrams, flowcharts: any visual aid that helps you explain what you think is happening with respect to the phenomenon you are studying can be a conceptual model.*

Variables & the relationships between them are the building blocks of your conceptual model.

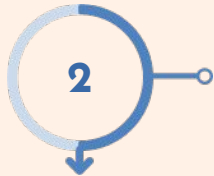


Conceptual models define the bounds of your work & help you frame your research question.



**Define the scope of your inquiry as clearly and narrowly as possible**

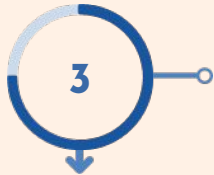
*Think of the temporal, spatial, and conceptual scope of your work. Why just this much? Logistical reasons, or a gap in literature?*



**Extract the key variables and constructs you are working with and arrange them in relation to each other.**

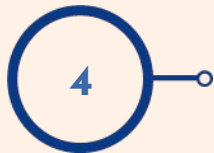
*Identify the IV, DV, mediating variables and moderating variables. Do you want to control for anything? What is the strength and direction of these relationships?*

*How will you operationalise them to measure them through your research? What does the literature say? Are there gold standard survey questionnaires you plan to use?*



**Explain why you have set up the conceptual model that way you have using theory and literature.**

*What are your assumptions for this model? When does this model not work? What are its limitations? Is there any variable you know affects this model but you are ignoring it? Why so?*



**Now draw your diagram (dashed arrows signify either weak links, or your hypotheses yet to be proven)**

*Go back to your research question: does this model help you answer it? Does your data collection tool and methodology address each part of this model? How so?*

# Anchor yourself to existing knowledge before you begin your work.

## Ensure you are telling a story

*Have you linked your review back to your research objectives? Ask: "Would a reader understand the current state of knowledge, & why my study is needed?"*

## Your review is an argument, not a list

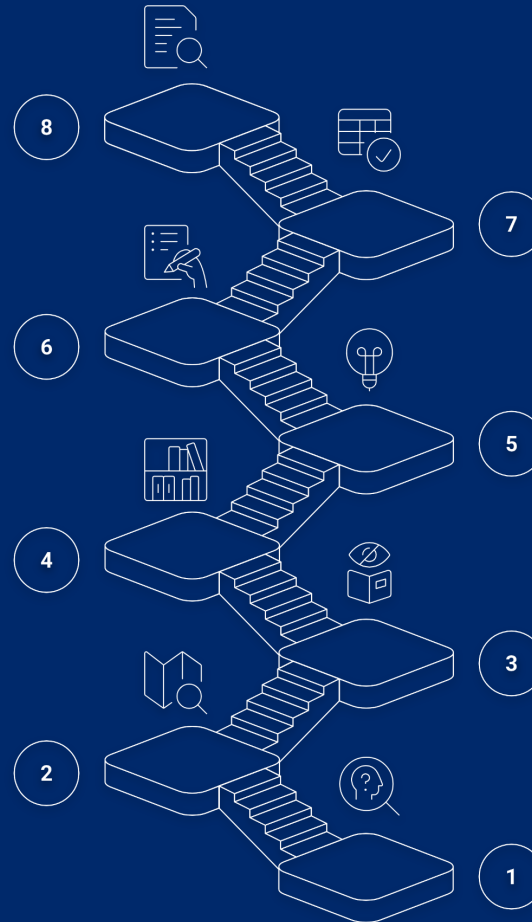
*Structure it clearly and logically, with an introduction, thematic sections, critical analysis of the selected papers, and a conclusion section that leads to your research question at hand.*

## Organise literature thematically

*Literature is supposed to help you justify your argument. Structure your review along themes, methodologies, chronology, & theoretical schools of thought so you can argue better.*

## Search strategically to help you find your way

*Use multiple databases (Wikipedia does not count). Backward & forward tracking helps you find papers in the same paradigm.*



## Be methodologically transparent & reference properly

*Explicitly record how you searched for, organised, & analysed the literature you reviewed to aid replicability. Use a consistent citation style & add footnotes and in-text citations where required.*

## Bring in your voice to synthesise & evaluate

*How do findings from various studies align or differ? What directions has the field taken over the years? What remains underexplored or methodologically weak? How does your study address these gaps? Make your case here.*

## Read critically to identify patterns, gaps, & contradictions

*What was the research question & methodology? What are the key findings? What are the limitations or biases? How does it connect or contrast with other studies?*

## Define your purpose & scope

*Clarify: what are you trying to understand or explain? Set spatial, temporal, and theoretical bounds for yourself.*

## Use the literature, don't let it use you.

Existing literature should be an oar in your hand, not a yoke on your back: treat it as a source of ideas and alternative ways of framing the issue you seek to explore.

Every paper and journal article you read is drawing from some **research paradigm**, even if it is not explicitly mentioned. Every paradigm has a way of making meaning of the world which give everyone using the paradigm some common ground to understand each other and “rules” to play the game.

Thus, when reviewing literature, ask yourself:



*What are the set of assumptions, methodologies, and preferences within which your work is grounded?*



*Why are you choosing this particular approach over others? How does it serve to answer your research question better than other approaches?*

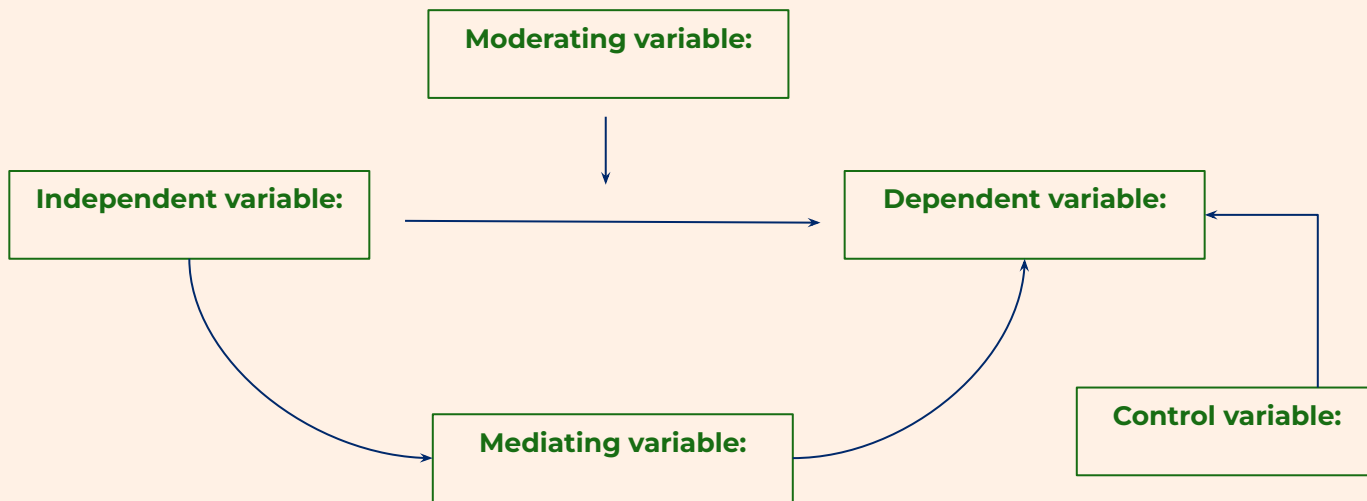


*What are the limitations of this approach? Are there other paradigms that would answer this question differently? Despite knowing this, why are you choosing this particular paradigm for your study?*



## Exercise

Create your own conceptual model using this template.



# Recommended readings

1. “Conceptual Models.” *NCPI - Student Assessment Toolkit*, Stanford. [URL](#).
2. “Conceptual model | Research Starters” *EBSCO Research*. [URL](#).
3. “How to Use Conceptual Models?” *SERC Carleton*, [URL](#).
4. Magher, Maria. “How to Make a Conceptual Framework for a Thesis” *seattlepi*, [URL](#).
5. “What is a Conceptual Framework?” *Scribbr*. [URL](#).
6. “Why Use Conceptual Models?” *SERC Carleton*, [URL](#).

## Output from this module

Create a conceptual model for your research question. You will add this to the background and rationale section of your IRC inception report.

## **MODULE THREE: DEFINING YOUR RESEARCH SCOPE**

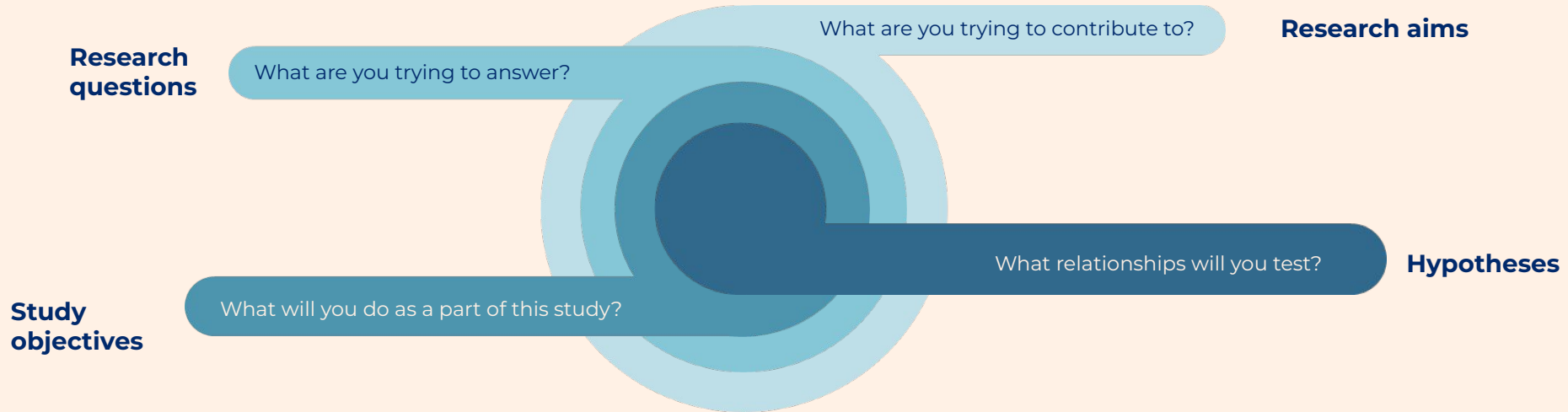
*What Will You Focus On, What Will You Leave Out?*

Are research aims, questions, hypotheses, and objectives all the same thing?

**No.**

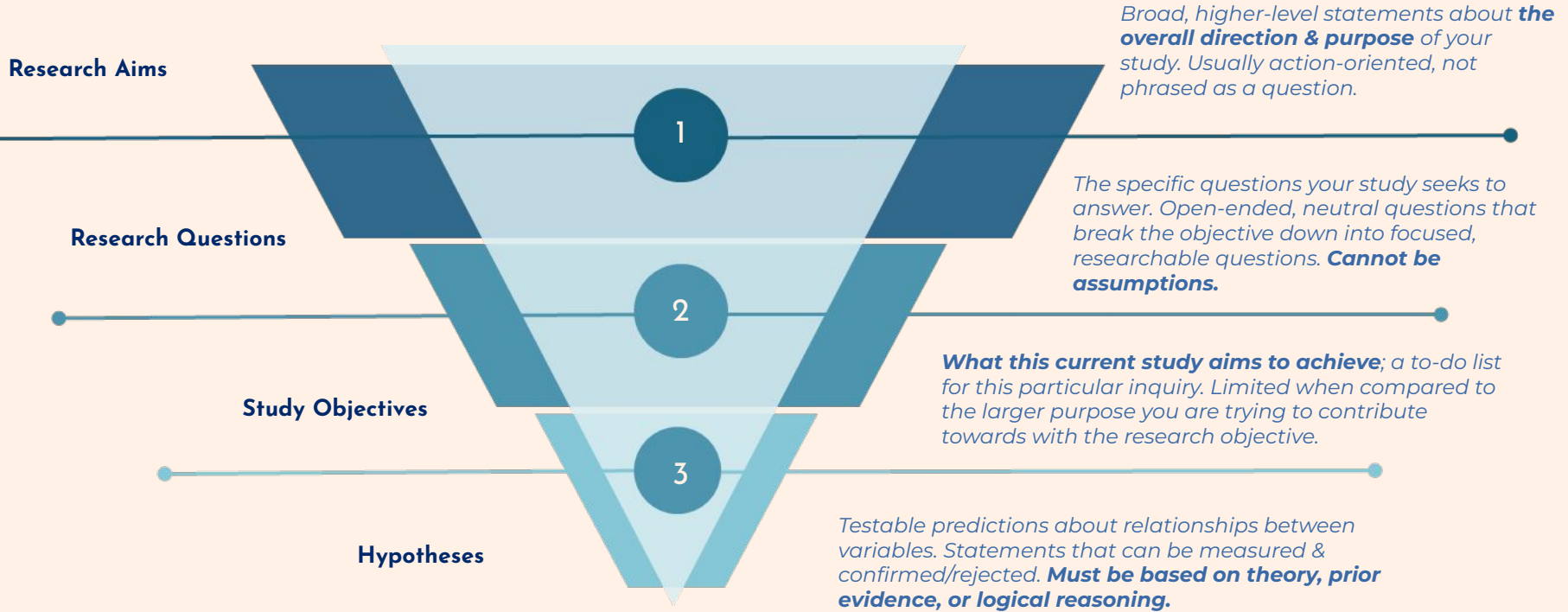
These are components help you systematically set up the scope of your enquiry.

You must go from higher-level boundary setting to granular to-do lists for your research with each step.



All four components must align to form the golden thread that unites your focus and scope, because **every decision you make in the research process will be informed by it.**

Each component serves a distinct purpose for your research study.



(Depending on the type and purpose of your study, you may or may not have hypotheses to test.)

Ambiguity is your enemy: write clear, well-defined objectives, questions & hypotheses.

|                       |                    | AVOID                                                                                                                                                                                    |
|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUANTITATIVE RESEARCH | Research question  | <ol style="list-style-type: none"><li>1. Vague &amp; unfocused questions</li><li>2. Closed questions (Y/N type)</li><li>3. Questions requiring a simple choice</li></ol>                 |
|                       | Research objective | <ol style="list-style-type: none"><li>1. Poor understanding of the research question and aims</li><li>2. Insufficient description of population, variables, or study outcomes</li></ol>  |
|                       | Hypotheses         | <ol style="list-style-type: none"><li>1. Unverifiable hypotheses</li><li>2. Incompletely stated groups of comparison</li><li>3. Insufficiently described variables or outcomes</li></ol> |
| QUALITATIVE RESEARCH  | Research question  | <ol style="list-style-type: none"><li>1. Ambiguous or oversimplistic questions</li><li>2. Questions unverifiable by data collection and analysis</li></ol>                               |
|                       | Research objective | <ol style="list-style-type: none"><li>1. Statements unrelated to the research question and aims</li><li>2. Unattainable or unexplorable objectives</li></ol>                             |
|                       | Hypotheses         | <ol style="list-style-type: none"><li>1. Statements simply expressing facts</li><li>2. Insufficiently described concepts or variables</li></ol>                                          |

([Barroga & Matanguihan, 2022](#))

Remember: framing good research questions and hypotheses is not only about defining what you will be studying (and how), but also about specifying what is **outside** the scope of your inquiry.



Your research goal determines the type of research question & hypotheses you write.

Do you want to:

1

Explore the relationship  
between two variables?

2

Predict a score or the  
membership of a group?

3

Test for differences between  
groups or treatment conditions?



These require relationship-based questions

This requires comparative research questions

Your research question needs to be bounded by time, space, and concepts.

Decide on the limits to your research question on these three parameters:

## Spatial

**What is the geographical or regional scope of your research study?**

*Is it limited to a town/city, block, district, or certain state(s)? Is it representative of all of India?*

*Why is this spatial limit necessary? How do you justify not going beyond this boundary for this research study?*

## Temporal

**What is the time period that your research study will consider?**

*Are you looking at a limited period in time, or starting from a certain year till the present?  
Is it a cross-sectional study?*

*Why is this temporal limit necessary? How do you justify not going beyond this boundary for this research study?*

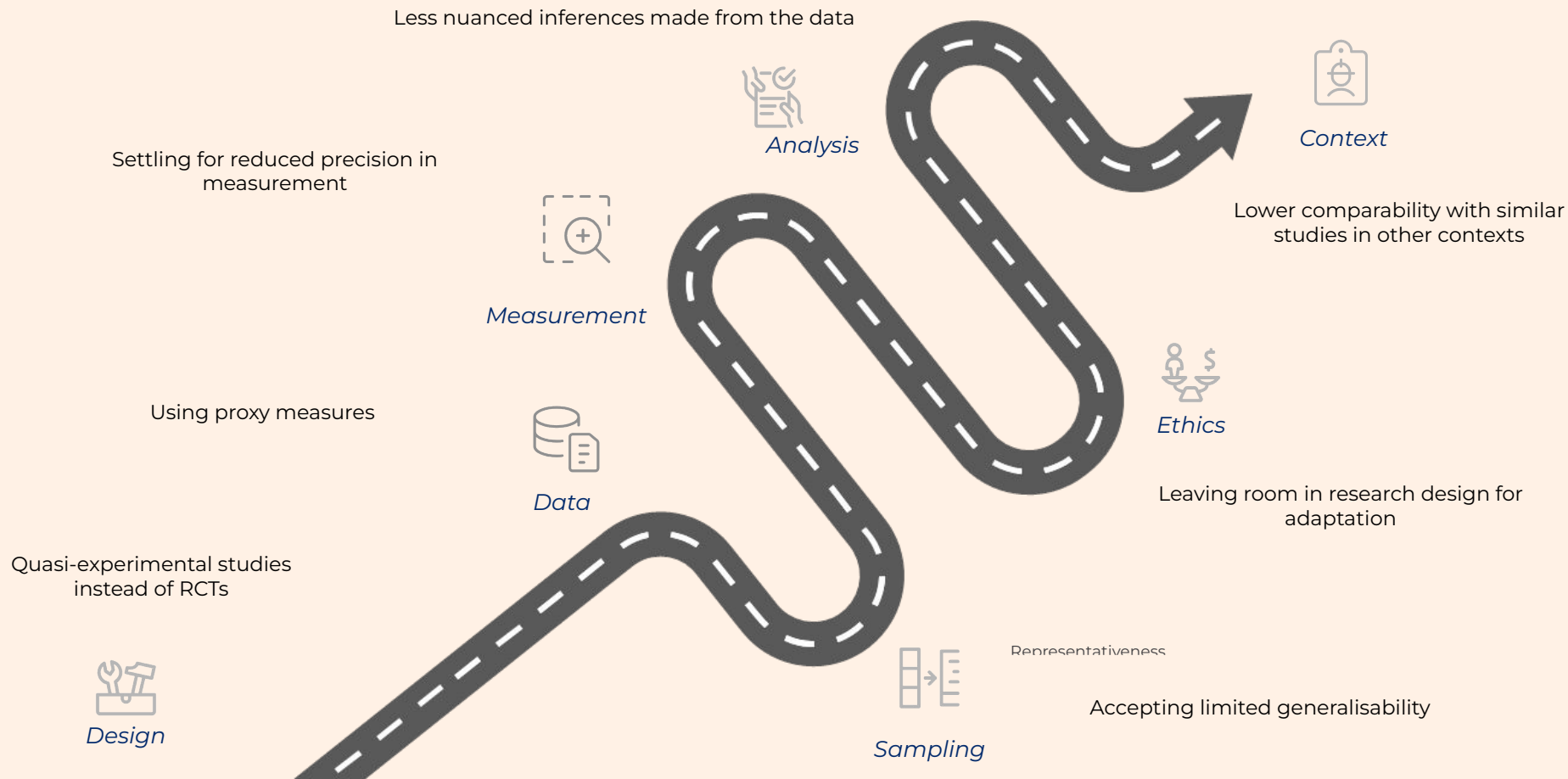
## Conceptual

**What concepts will you be limiting yourself to in your research study?**

*Are there only certain constructs and variables you will be focusing on as a part of the concepts you operationalise? Which ones will you be leaving out and why?*

*Why is this conceptual limit necessary? How do you justify not going beyond this boundary for this research study?*

The scientific method typically has a cost to it, so you will have to balance rigour and feasibility.



# Exercise

*Go back to your research question and evaluate it critically for the following:*

1. Do you have separate research aims, research questions, study objectives, and hypotheses (if applicable)?
2. Are each of these components well articulated?
3. Have you defined the spatial, temporal, and conceptual scope for each of your research questions?
4. What are the trade-offs you think you will be making between rigour and feasibility at this level? Can you begin thinking of ways to address them through some choices you can make when you pick your research methods?

# Recommended readings

1. Campbell, D. T., & Stanley, J. C. (1963). *Experimental and Quasi-Experimental Designs for Research*. Boston: Houghton Mifflin. [URL](#).
2. Grad Coach (n.d.). *Research Questions, Objectives & Aims (+ Examples)*. Grad Coach Research Guides. [URL](#).
3. K. Sarajlic, S. Pranjic, A. Alibegovic (2022). *A Practical Guide to Writing Quantitative and Qualitative Research*. *Acta Informatica Medica*, 30(2): 145–150. [URL](#).
4. Laerd Dissertation (n.d.). *Deciding whether to include research questions and/or hypotheses*. Laerd Dissertation Guides. [URL](#).
5. Morgan, D. L. (2007). *Paradigms lost and pragmatism regained: Methodological implications of combining qualitative and quantitative methods*. *Journal of Mixed Methods Research*, 1(1), 48–76. [URL](#).
6. Ohio State University (n.d.). *Research Questions & Hypotheses | Quantitative Methodology Center*. Quantitative Methodology Center. [URL](#).
7. Shadish, W. R., Cook, T. D., & Campbell, D. T. (2002). *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Boston: Houghton Mifflin. [URL](#).

## Output from this module

Finalise the research question after refining it using the rules of thumb you have learnt in this module. Present your research question (and any sub-questions) as a concise statement.

Write 1-2 pages to delineate accompany your research question, which marks the temporal, spatial, and conceptual scope for this study, and provide justifications for this scope you have set for yourself.

## **MODULE FOUR: DECIDING ON THE RIGHT RESEARCH METHODOLOGY**

*How Will You Answer Your Research Question?*

Research methodology is the overall strategy you use to address research questions.

This includes **research philosophy, design,** and **methods** you will use.

*What is the nature of reality and knowledge?*

Your answer to this question will decide your research philosophy, and the subsequent methods you will use to answer your research question.

Some **common paradigms/philosophies used in the social scientific world** are as follows >>



### Positivism

- A single, objective, measurable reality
- Quantitative methods to find objective truths
- Explains & predicts phenomena



### Post-positivism

- Reality exists, but it can only be approximated
- Mixed methods that stay close to positivism
- Understands reality through rigorous testing



### Critical Theory

- Reality is shaped by power structures & inequalities
- Qualitative methods to critique social structures
- Critiques and challenges existing power dynamics to create social change



### Interpretivism

- Multiple realities exist, subjective & socially constructed
- Qualitative methods to understand subjective experiences
- Understands & interprets meaning of social phenomena from subjects' perspective



Weave the 'golden thread' in this stage for methodological coherence.

**Research design** is the **plan** for your study. **Research methods** are **techniques** you will use to execute that plan.

## Methodology

Provides the strategy to answer the question; informed by the conceptual framework. Hypothesized relationships tested with quantitative methods, lived experiences explored with qualitative ones.

## Data collection & analysis

Consistent with the methodology opted for; must measure key variables in the conceptual framework and must fit both the type of data collected, as well as the research question.

## Research questions

Form the foundation of the study; drives the overall design. Determine types of data needed & what needs to be explored.

## Conceptual framework

Informs what you study & how; translates the question into a visual/logical structure that justifies why relationships exist. Guides what needs to be measured.



The kind of research question you ask determines the methodology you use.



**Descriptive questions**  
where you want to  
quantify trends and  
frequencies will require  
**quantitative approaches.**



**Evaluative questions**  
where you want to assess  
both outcomes and  
processes will require  
**mixed methods  
approaches.**



**Exploratory questions**  
where you want to  
understand meanings and  
experiences will require  
**qualitative approaches.**



**Explanatory/causal  
questions** where you want  
to produce causal  
explanations for  
phenomena will require  
**experimental or  
quasi-experimental  
approaches.**



**Predictive questions**  
where you want to  
forecast future outcomes  
or behaviors based on  
current and historical data  
will require **modelling  
approaches.**

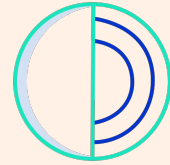
Your research design is determined by what you want to track and how.



**Experimental designs**  
such as RCTs help  
establish causality.



**Quasi-experimental designs**  
use comparison groups  
without randomisation.



**Cross-sectional studies**  
describe trends or  
characteristics at a point in  
time.



**Longitudinal studies**  
track changes over  
time (time series or  
panel).

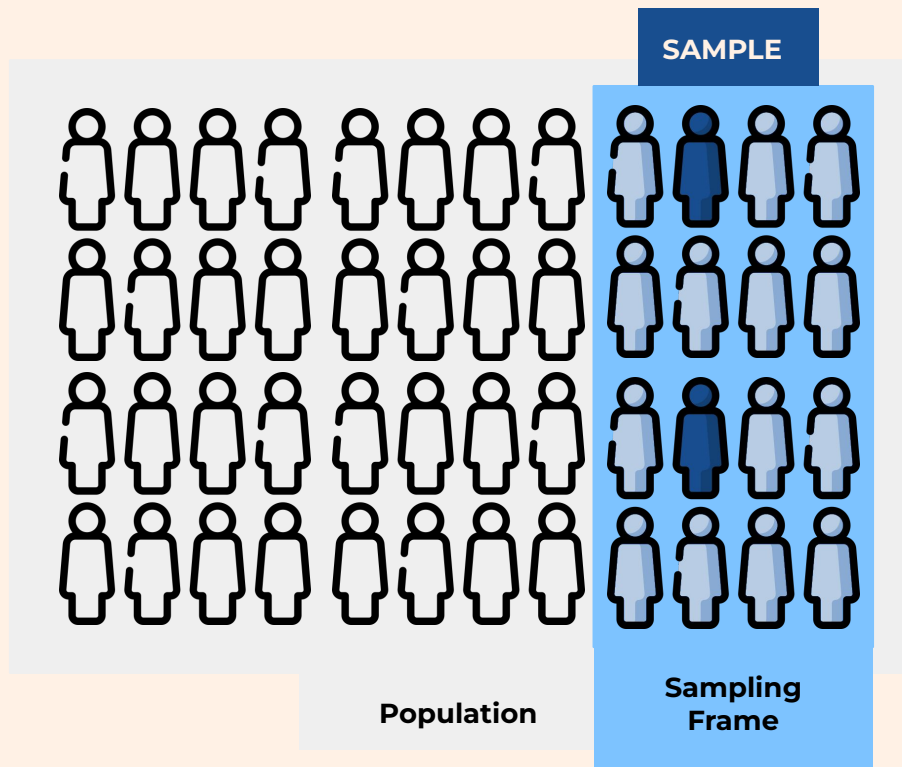


**Case studies** examine  
a single or multiple  
contexts closely.



**Action research** uses  
a participatory and  
iterative approach for  
real-world settings.

Truly random sampling is easier said than done.



- Simple random sampling requires you to have a complete and accurate list of the population: which is almost never possible.
- Almost always, your sampling will be **purposive** to some extent: you will strategise to ensure representation.

The **population** is the entire universe of units you want to analyse; the **sampling frame** is the realistic subset you actually can access.

Your aim in a typical quantitative study, however, is to sample in a way that your findings are generalisable to the entire population.

## Quantitative approaches use probability sampling methods for generalisability.

Since our intent is to say something about the larger population based on a small sample, sampling for such an inquiry is **representative** of the population.

*You will most likely use one of these four strategies in most cases:*

### Simple random

Each member of the population has an equal chance of being selected

### Stratified random

Population is divided into subgroups (strata), and a random sample is drawn from each of these strata

### Systematic

A random sample is drawn from the population by selecting units at regular intervals starting from a random point

### Cluster

Groups (or clusters), rather than individual units (or people) of the target population are selected at random for the sample.

# There are a lot more sampling strategies for qualitative data collection than you'd think.

Since our intent is to highlight the particular and the specific, sampling for such an inquiry is mostly **purposive**.

*You will most likely use one of these four strategies in most cases:*

## Random/stratified purposive

Adds credibility, facilitates subgroup comparison

## Maximum variation

Identifies a wide range of diverse individuals/sites

## Snowball/chain

Identifies cases from people who know which would be info-rich

## Convenience

Saves time, money, effort at the expense of information, credibility

*...but these are not the only ones available to you.*

| Sample type                    | Definition                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------|
| Homogenous                     | Focuses, reduces, simplifies.                                                       |
| Critical case                  | Permits logical generalisations that can be applied to other cases                  |
| Theory-based                   | Elaborates on/examines a construct of a theory/entire theory                        |
| Confirming/disconfirming cases | Elaborates on initial analysis, seeks exceptions, looks for variations              |
| Extreme/deviant case           | Learns from highly unusual information manifestations of the phenomenon of interest |
| Typical case                   | Highlights what is normal or average                                                |
| Criterion                      | Seeks cases that meet some criterion                                                |

See [Miles & Huberman \(1994\)](#) for a full list

# Modelling is an ideal approach for when you want to answer predictive questions.

Modelling helps us understand systems that are difficult to observe directly, especially when:



Relationships and processes are complex, dynamic, or nonlinear



Interventions have indirect effects that unfold over time



Experiments are impractical, too costly, or ethically constrained



We want to simulate "what if" scenarios before acting in the real world

Modelling is not a replacement for empirical research — it extends and complements it, offering a structured way to think about systems.

# A model is a simplified representation of a real-world question, built to answer a precise question.

## A good model:



## Models range from:



Conceptual models (flows, diagrams, causal loops)



Statistical models (regression, classification, forecasting)



Spreadsheet models (simple arithmetic or decision tools)



Dynamic models (STELLA, system dynamics, hydrological models)



Simulation models (agent-based, scenario modelling)



# Different research questions will require different types of modelling approaches.

Matching modelling approaches to types of research questions is important. Some examples are as follows:

## Causal/Explanatory

- What mechanisms drive groundwater decline in Raichur?
- How does water availability affect crop choice and income over time?

## Predictive

- What will happen to the water table if irrigation demand increases by 20%?
- How will urban water supply change under different rainfall futures?

## Exploratory/Scenario-based

- What are tradeoffs between short-term parameters (e.g. water security, income) and long-term parameters (e.g. soil-health, productivity) for farmers
- What are the trade-offs between energy subsidies and groundwater sustainability?

If your question requires time, dynamics, policy trade-offs, feedback loops, or multi-variable interactions, modelling is almost always appropriate.

## Models built without grounding in lived realities quickly become misleading.

Models need reality checks. Before building a model, you must understand the system empirically:

- Define the **system boundaries** clearly
- Map **key variables** (stocks, flows, feedbacks, constraints)
- Identify **available datasets** (administrative, satellite, survey, local knowledge)
- Use fieldwork & qualitative insights to **shape assumptions and hypotheses**
- **Cross-check parameters** with domain experts, implementers, and community actors

This is where organisations like WELL Labs are strong (and are training you too):

***Strong field presence + robust data + modelling → meaningful and realistic insights***

## Always sense-check your model.

A model should be judged by its usefulness, transparency, and predictive accuracy, not its complexity.

Good modelling practice requires you to:

### Validate

**Compare outputs with  
real-world historical data.**

See if the model replicates  
observed patterns

*Ask: Does this make sense  
based on what we know?*

### Triangulate

**Use multiple data sources  
(rainfall, crop cutting, energy  
data, farmer interviews).**

Bring in sectoral experts to  
pressure-test assumptions.

*Compare with other models,  
benchmarks, or studies*

### Iterate

**Update and refine  
assumptions as evidence  
changes**

Document all changes and  
justify them

*Conduct sensitivity analysis  
(which variables matter most?)*

Remember George Box's quote: ***"All models are wrong, some are useful."***

## Methodology

Mixed-methods design — because the research question includes both causal (how participation influences productivity) and process-oriented (how participation happens) aspects.

Quantitative component: to measure impact on crop yield, income, and water table.

Qualitative component: to understand community perceptions, inclusiveness, and sustainability practices.

## Research questions

How does community participation in watershed management projects influence agricultural productivity and household livelihoods in semi-arid regions of Maharashtra?

## Data collection & analysis

### **Data collection:**

Quantitative: Household surveys with structured questionnaires (e.g., sample of 300 farmers across 6 villages).

Qualitative: Focus Group Discussions (FGDs) and Key Informant Interviews (KIIs) with project staff, self-help group members, and panchayat representatives.

Secondary data: rainfall, soil moisture, and productivity data from government sources.

### **Data analysis:**

Quantitative: Statistical analysis — regression to test relationship between participation levels and productivity outcomes.

Qualitative: Thematic analysis of FGDs to identify patterns in local governance, inclusion, and perceptions of benefits.

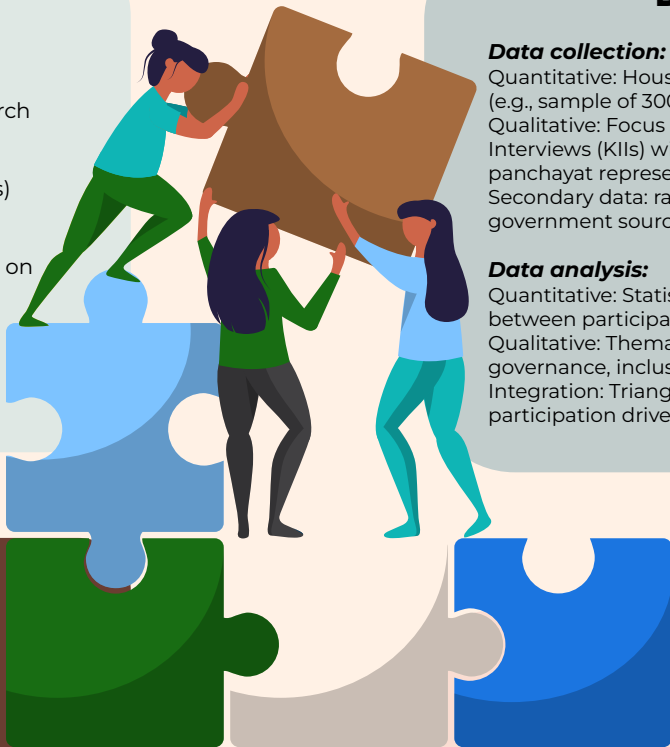
Integration: Triangulate findings to explain how and why participation drives outcomes.

## Conceptual framework

Based on Participatory Development Theory and Sustainable Livelihoods Framework:

Participation → better maintenance of watershed structures, improved water availability → higher agricultural productivity → enhanced livelihoods.

Moderating factors: institutional support, social cohesion, access to markets.



# Exercise 1

Use this research design matrix to link your questions to the methods you will use.

| <b>What</b> do I need to know? | <b>Why</b> do I need to know this? | <b>What kind of data</b> will answer these questions? | <b>Where</b> can I find the data? | <b>Whom</b> do I contact for access? | <b>Timelines</b> for acquisition |
|--------------------------------|------------------------------------|-------------------------------------------------------|-----------------------------------|--------------------------------------|----------------------------------|
|                                |                                    |                                                       |                                   |                                      |                                  |
|                                |                                    |                                                       |                                   |                                      |                                  |
|                                |                                    |                                                       |                                   |                                      |                                  |
|                                |                                    |                                                       |                                   |                                      |                                  |

## Exercise 2

Build a mini system model: develop a simple conceptual or numerical model of a system relevant to your research question.

Your **outputs** should include:

1. Research question the model answers
2. Variables included (stocks, flows, controls)
3. Causal diagram or simple equation-based structure
4. Assumptions (clearly listed)
5. Where you would get the data to parametrize the model
6. One scenario you would test
7. What you would use to validate the model

This does not need software — a hand-drawn diagram or spreadsheet is fine. The goal is to learn to think like a modeller.

# Recommended readings

1. Creswell, J. W. (2014). *Research design: Qualitative, quantitative, and mixed methods approaches* (4th ed.). SAGE Publications. [URL](#).
2. Tashakkori, A., Johnson, R. B., & Teddlie, C. (2020). *Foundations of mixed methods research: Integrating quantitative and qualitative approaches in the social and behavioral sciences* (2nd ed.). SAGE Publications. [URL](#).
3. Olsen, W. (2022). *Systematic mixed-methods research for social scientists*. Springer Nature. [URL](#).
4. Fetters, M. D. (2023). *The mixed methods research workbook: Activities for designing, implementing, and publishing projects*. SAGE Publications. [URL](#).
5. Poth, C. N. (Ed.). (2023). *The SAGE handbook of mixed methods research design*. SAGE Publications. [URL](#).
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## Output from this module

Draft a section on research methodology for your inception report.

Connect your research questions, conceptual framework, methods, and research design together in this section so you justify your choices before proceeding to lay out your research plan in the next section of the report.



## **MODULE FIVE: DATA COLLECTION AND FIELDWORK**

*How Will You Actually Collect the Data You Need?*

## Standardising the questions you ask in a survey ensure validity and reliability.

The questionnaire is called the **survey tool**. Each individual question on the questionnaire is called a **survey item**.

**Gold standard survey tools** are usually used to create questionnaires that are valid.

Different kinds of response options exist on quantitative surveys:



Numerical



Categorical



Multi-select/  
rank choice



Open text



Likert scale

Qualitative research most often uses in-depth interviews and focus-group discussions, which require a **semi-structured interview guide**. Instead of response options, these guides use probes & follow-up questions to aid the data collection.

**Piloting** your tools before launching the survey helps you make changes to the questions as required.

You can also use this time to test different versions of your survey items (A/B testing).

## Guard against bias during sampling, tool design, & data collection.

**Bias** can be introduced by researchers or the research design:

- Sampling bias
- Question order bias
- Confirmation bias
- Observer bias
- Non-response
- Hidden bias



**Ways to address common types of bias** include:

- **Sample representatively**, and **use peer review** throughout the research design process to minimise researcher bias
- Word survey tools **neutrally and clearly**; add **sanity check items** and randomised question order where possible. Gold standard tools are useful for this
- **Anonymity** helps with reducing respondent bias
- For bias that cannot be addressed during data collection, **weights** can be assigned during statistical analysis

...or displayed by **respondents**:

- Social desirability bias
- Halo effect
- Cognitive bias
- Recall bias
- Demand bias



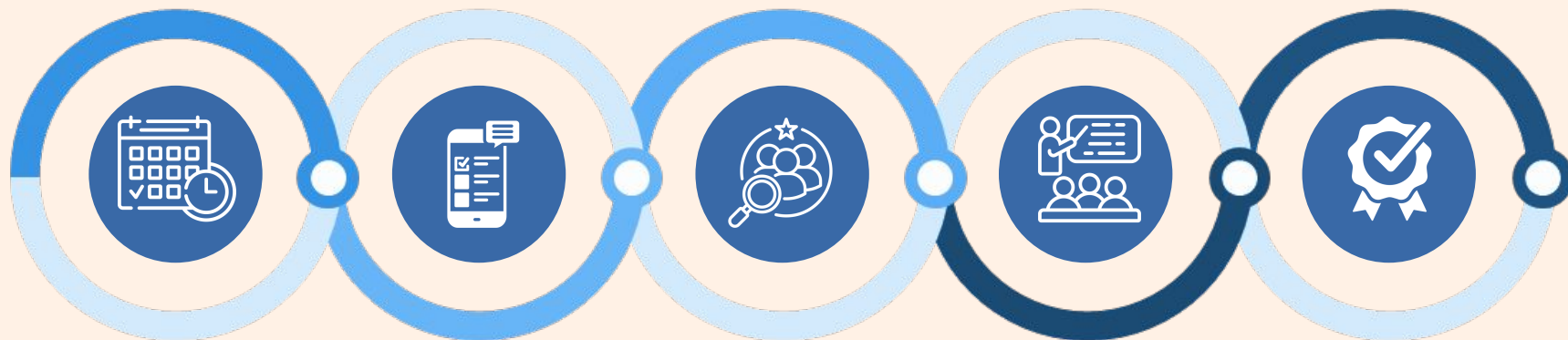
# Good primary data collection requires careful planning and execution.

- Daily surveyor productivity
- Survey team size
- sampling design
- Buffer time

– all play a role in designing your data collection schedule.

Recruiting and training surveyor teams can be done through an implementation partner or through contract surveyors.

High frequency checks, back checks, and real-time monitoring of indicators of interest will help in data quality control.



Use CAPI software like KoboCollect, SurveyCTO, mWater etc. to program your surveys.

These make quality assurance and version control easier.

Training content should broadly cover:

- Org policies
- Study overview
- Team structure
- Surveying & field protocols\*
- Survey tool walkthrough
- Mock surveys/field practice

\*Informed consent, research ethics, back checks and data quality control, field locations and respondent selection, de-escalation.

**Attrition** is an important consideration: especially in longitudinal studies.

Typical attrition rates in field experiments are estimated at 10-15%. Account for this in your sample size and data collection plan.

## Research ethics go beyond addressing bias.

Most academic research requires **ethics approval** to ensure there is no inadvertent harm done to participants.

You have **a duty of care** towards your participants. The three pillars of ethical human subjects research:



### **Informed consent is key.**

*Communicating project background, voluntary participation, and data confidentiality to are usually mandatory.*

*Participants should have the opportunity to ask questions before, during, and after the survey.*



As inconvenient as it might seem, **voluntary participation is the respondents' right.**

*They have a right to not answer any question they may not wish to, as well as a right to withdrawing their data either partially or completely from the study even after the survey completion.*



### **Data confidentiality is your responsibility.**

*This includes removing personal identifiable information (PII) so that data is anonymised, especially if it is to be published in the public domain.*

*Aggregating data is an important step in making data publicly available as well.*

In India, caste, gender, class, and other intersectionalities affect your fieldwork.

### COMFORT

Based on the local customs, the process of establishing rapport can be different. If you're the way you open the interview is friendly and relaxed, respondents will be encouraged to speak with you.

### INFORMED CONSENT

It is important to explain all the important information about the study to them properly so their most important doubts are answered. Having a script, consent form, and info sheet ready helps.

### LANGUAGE

Never use words and/or tone of voice to show we agree/disagree or approve/disapprove of anything the respondent is telling us. Remember, the interview is about their experience, not ours.

### TACT

Getting an honest answer to a sensitive topic is tricky: try asking such questions in this manner: "Some people say [XYZ]. What do you think?" Active listening can help cross-question any conflicting/ "perfect" answers.

### FEEDBACK

Subtly show that you have taken the time to understand how their field works (keep in mind what others has said to help with probes), and even during the interview you have been listening. Use phrases that help in back-checks & summary.

### RESPECT

Respondents are giving us their time and trust us with their experiences. It is thus our duty to respect the respondents and throughout the interaction treat them how we would like to be treated and spoken to.

# Engaging with your positionality and practising reflexivity makes you a better researcher.



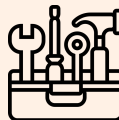
**Positionality** refers to the position a researcher has chosen to adopt within a given research study ([Savin-Baden & Major, 2013](#)). It requires you to acknowledge your own **caste, class, gender, and other realities** that affect how you understand and engage with the work you do.

**Reflexivity** involves addressing the issues of identity and positionality by making the researcher's assumptions explicit and finding strategies to question these ([Lazard & McAvoy, 2017](#)).



## At the conceptual level

- Be **flexible** with concepts & assumptions
- **Abduction** (thought experiments and best guesses)-but pair this with induction and deduction



## At the methodological level

- **Triangulate** your findings through different kinds of methods, respondents, researchers
- Focus the right amount on details and context: there is such a thing as **too much detail**
- Actively look for **contradictions** and deviant cases that prove you wrong



## At the reflexive level

- **Personal reflexivity**: confessional vs realist tales
- **Methodological reflexivity**: document how you went from intuition to accountable public knowledge, establish an audit trail of your procedures and choices
- **Theoretical reflexivity**: be open about your orientations, biases, blind spots



# Outputs from this module

Create a **week-wise timeline** (gantt chart) for your research project. Ensure that this timeline is created based on logistical considerations that you anticipate will be required to collect data for your project.

Write a section on the **anticipated risks** (threats to validity, logistical issues, possible bias and others) and how you plan to address the same. Also list out all **major assumptions** made by you throughout the research design and planning stage.

Attach the **final survey questionnaire** (for primary data collection) and/or the data collection plan (for secondary data collection). For survey tools, also mention which existing questionnaires you are using as gold standard.

## **MODULE SIX: ANALYSING DATA AND REPORTING YOUR FINDINGS**

*How Will You Make Sense of the Data You Collect?*

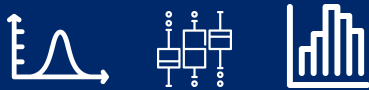
# Your research method determines how you analyse your data.

Connecting once again to your research question(s):

- Do you want to know **what/how much**—or trends and patterns in the data?
- Do you want to know **why/how**—to answer questions about processes, perceptions, and behaviours?

**Quantitative analysis** consists of using:

Descriptive statistics to describe the characteristics of a sample using **measures of central tendency**.



Represented commonly through frequency distributions, percentages, cross-tabs. Histograms, bar charts, boxplots, and scatter plots are used to visualise.

Inferential statistics to say generalise and predict for the larger population through the sample drawn.



Usually done through hypothesis testing (t-tests, chi-square, ANOVA, correlation, regression). This is where **p-values** and **statistical significance** are important.

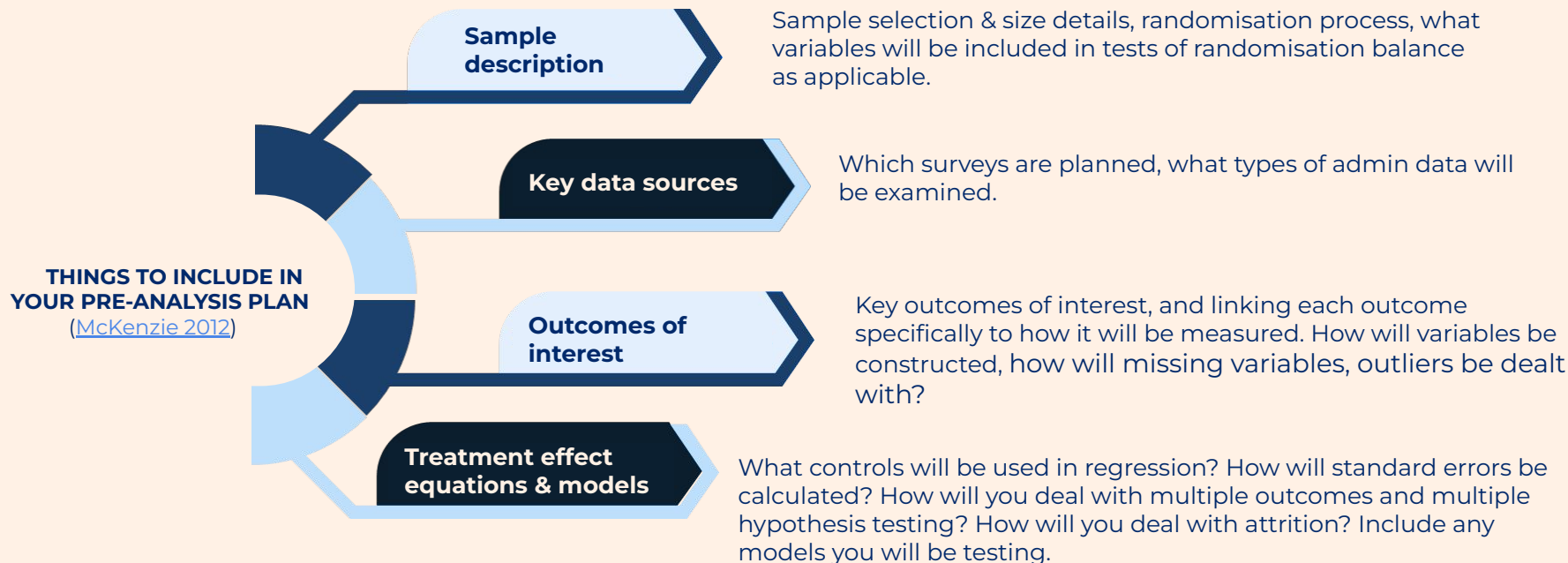
**Quantitative analysis** consists of multiple methods to make sense of textual data

- Coding (open or *a priori*)
- Thematic analysis
- Narrative synthesis
- Other methods like archival research can help support interpretations.



## Plan beforehand what you will do with the collected data.

To address confirmation bias, especially in quantitative research, it is advised that you create a **pre-analysis plan** that lays out which relationships you will be testing, and what variables you will be using to test each of those once all the data is collected.



Once you have executed your pre-analysis plan, you may conduct **exploratory analysis** of your data based on the emerging relationships between variables and your time in the field.

# Output from this module

Create **a pre-analysis plan** based on your research questions and hypotheses.

Document systematically which variables you will use to answer each question, what statistical tests you will perform on them, and what data visualisations you think are most apt to communicate findings for each question.

Research communications as a separate module drafted along with Comms:

Storytelling with data (Pavan's upskilling session)

**Effective science communication** (section on writing with Ananya, forthcoming will be done by Dec)

(This can come later in the process, where the students start writing their outputs.)